

Lesson1

- 4 The sum $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{2021}{2022!}$ can be expressed as $a - \frac{1}{b!}$, where a and b are positive integers. What is $a + b$? (2014 AMC 10A Problems, Question #9)

A. 2020 B. 2021 C. 2022 D. 2023 E. 2024

- 6 Calculate: $\frac{4}{3 \times 7} + \frac{4}{7 \times 11} + \frac{1}{11 \times 12} + \frac{3}{12 \times 15} + \frac{2}{15 \times 17} + \frac{12}{17 \times 29} + \frac{1}{29} = \underline{\hspace{2cm}}$.

- 7 Calculate: $\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \frac{1}{63} + \frac{1}{99} + \frac{1}{143} = \underline{\hspace{2cm}}$.

Lesson 2 extention

- 5 If a four-digit number $\overline{2X9Y} = 2^X \cdot 9^Y$, then $X = \underline{\hspace{1cm}}$ and $Y = \underline{\hspace{1cm}}$.
- 7 Among the natural numbers less than 5000, how many numbers are divisible by 11 and have a digit sum of 13?
- A. 15 B. 18 C. 21 D. 24

Lesson 3

- 10 The least common multiple of x and y is 60. The least common multiple of y and z is 70. The least common multiple of x and z is 84. Then $x = \underline{\hspace{1cm}}$.

A. 6 B. 10 C. 12 D. 15 E. 20

Extention

- 9 Let $a@b = [a, b] + (a, b)$, where $[a, b]$ represents the least common multiple of a and b , and (a, b) represents the greatest common divisor of a and b . Given $12@x = 52$, then $x = \underline{\hspace{1cm}}$.

Lesson 4

If x and y satisfy the system of linear equations $\begin{cases} 323x + 457y = 1103 \\ 177x + 543y = 897 \end{cases}$, find the value of $x^4 + 4x^2y^2 + 5y^4$.

L4 Ext

9 Solve the following system of linear equations:

$$\begin{cases} |x| + y = 12 \\ x + |y| = 6 \end{cases}$$

Lesson 5

1 Solve $|x - 2| \leq 2x - 10$.

2 Solve $|2x - 4| + |3x - 3| \leq 5$.

Lesson 5 HW

- 5 Solve the following system of linear inequalities $\begin{cases} \frac{x-3}{2} + 3 > x+1 \\ 1-3(x-1) \leq 8-x \end{cases}$, and graph the solution.

Lesson 5 Ext

- 10 Solve the following absolute-value inequalities
(1) $|3x + 1| > 2x$ (

$$(1) |2x - 3| \geq x + 1$$

10 Solve the following absolute-value inequalities:

$$(1) |2x + 1| - |5 - x| > 2$$

$$(2) |x - 1| + |x + 1| < 9$$

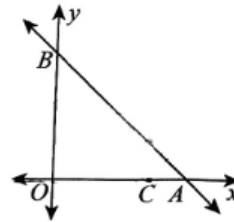
L7

- 8 If the graph of the line $y = 2x - 1$ is rotated 90° counterclockwise about the origin O , the equation of the line after the rotation is _____ .

L7 Extention

- 8 The equation of the line obtained by rotating line $y = 3x$ clockwise around the origin 90° is _____ ; The equation of the line obtained by rotating line $y = 3x$ counterclockwise around the origin 90° is _____ .

- 10 The graph of the linear equation $y = -x + 5$ intersects with x -axis and y -axis at A and B respectively .



- (3) Points M and N are on the line AB and y -axis respectively, so that the perimeter of $\triangle CMN$ is at its minimum. What are the coordinates of point N ?

L8
3

$$(3) (2a + b)^2 [4a^2 - (2a - b)b]^2 = \underline{\hspace{2cm}}$$

Lesson 9

2 Factorize: $x^2(a+b)^2 - 2xy(a^2 - b^2) + y^2(a-b)^2$

L9 HW

8

(4) $-20x^2 + 9x + 20 = \underline{\hspace{2cm}}$.

L9 Extention

5 Factorize: $x^2 + 144y^2 - 25xy = \underline{\hspace{2cm}}$.

4 Factorize: $(2x - 3y)(3x - 2y) + (2y - 3x)(2x + 3y) - (6x - 4y)(6x - 9y) = \underline{\hspace{2cm}}$.

8 Factorize: $x^2 - (a + \frac{1}{a})x + 1 = \underline{\hspace{2cm}}$.

L10 Extention

- 7 Given n is a certain natural number in $1 \sim 2021$. If the 72 multiple of n is a perfect square, then the maximum value of n is _____ .

L11 extension

- 7 What is the value of a two-digit number which is 9 times the sum of its digits?

- 7 The sum of the following numbers in base-10 is _____ .

$$(214)_5 \quad (ABA)_{16} \quad (1234)_8$$

L16 Extention

- 7 If the quadratic equation $(k + 1)x^2 + 3x + 1 = 0$ in terms of x has two different solutions, the range of k is _____.

- 10 The sum of the reciprocals of the roots of the equation $2020x^2 + 2019x + 2018 = 0$ is _____. (Adapted from 2003 AMC 10A Problems, Question #18)

L17

- 5 The number 2013 has the property that its units digit is the sum of its other digits, that is $2 + 0 + 1 = 3$. How many integers are less than 2013 but greater than 1000 share this property?

A. 33 B. 34 C. 45 D. 46 E. 58

L18 Extention

- 9 DeepSeek software is a deeply exploratory artificial intelligence software developed in China. Now rearrange the letters of the word DeepSeek. The number of different arrangements where the letters e are not adjacent is _____ .

A. 24 B. 120 C. 576 D. 2880

- 10 A, B, C, D, E, F form a line, satisfying the conditions that A, B are adjacent, C, D are not adjacent, and E does not stand at the two ends. How many different arrangements are there?

A. 48 B. 96 C. 72 D. 144 E. 288

9 A team has 5 boys, 4 girls. Now we need to pick 3 people from this team to do the cleaning.

(1) If we are to pick at least 2 boys, there are _____ ways of picking.

(2) If we are to pick at least 1 girls, there are _____ ways of picking.